AI-Driven Speech-to-Text Systems for Enhanced Accessibility in the Hearing Impaired Community

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***Abstract*— Although today more and more assistive technologies are being developed, deaf and hard-of-hearing people still experience numerous communication difficulties both in school, the workplace, and in social settings. We assess how the current AI-based speech-to-text (STT) systems are performing on addressing these accessibility obstacles in this paper. To evaluate the performance of the 5 major Studio tunnel transmitters, we conducted a test of their performance in dissimilar acoustic settings and contrasted the results in terms of transcription accuracy, latency, and contextual accommodation. In our evaluation, we factored in the opinion of 127 hearing-impaired users who were put to the test on 1st-hand testing and controlled interviews. Findings show that indeed the accuracy of real-time transcription (87-94) is significantly better than the prior generation systems, and the ability to get specialized domain vocabulary appears to be an area of strength. However, there was complete impairment of performance in noisy conditions and non-standard speech. User satisfaction levels were associated with Transcription speed and not with absolute accuracy. These results indicate that though the existing AI-powered STT systems have a tremendous positive effect on accessibility, noise robustness, and speaker adaptation adjustments should be made so that no competitive communication gaps with the deaf and hard-of-hearing audience remain.**

Keywords—Artificial intelligence, hearing-impaired, speech-to-text, transcription.

I. Introduction

Communication is one of the key human rights, but to the 466 million deaf individuals around the world, there is a barrier to conversation in most places, regardless of technology. The deaf and hard-of-hearing population has long been victims of systematic exclusion from much of the social, educational, and work life, and this has led to historically existing differences in quality of life and access to economic resources, which further result in inequality [1]. The best solution to an area of correction of such inequities is through improvement in assistive technology, specifically in speech-to-text (STT) devices, which are used to convert audio inputs into written messages in real-time. The traditional assistive technology, like the wearing of a cochlear implant to facilitate hearing, has not proved effective for the most challenging acoustic conditions and has left vast communication gaps among the people who have embraced the application of the assistive interventions. The introduction of artificial intelligence has rapidly led to the development of STT systems that can be considered a potentially revolutionary technology that can potentially change the lives of people with hearing disabilities by changing the way people communicate with each other in a variety of situations [2].

The combination of deep architecture models, computing power advancement, and enhanced training dataset sizes has brought enormous progress in automatic speech recognition systems over the last decade [3]. The modern AI-powered STT engines are currently showing the spectacular performance of up to transcription at the optimised mode of over 95% of clean speech patterns and optimally under favourable acoustic environments. These devices have been implemented on varying platforms that include standalone software and built-in operating system features, as well as specific hardware devices that would have varying benefits in diverse applications and environments [4].

In spite of these technological achievements, significant gaps remain between what is actually known about the usefulness of AI-powered STT technologies in meeting the real needs of the hearing-impaired members in actual communication scenarios. The current literature has focused largely on experimental settings, which have been assessed using measures of technical performance, whereby, in some instances, the lived realities of end-users were overlooked [5]. Technical tests also mainly focus on the rates of word mistakes and the accuracy figures, but do not give much information on usability in the real-world context in terms of perceived mental load, emotional status, and situation correctness [6]. Also, communication patterns and technological mastery. Furthermore, the ability of STT systems to work satisfactorily within a large variety of acoustic conditions, speaker populations and vocabularies of technical material have not been sufficiently reported, which casts doubts on their reliability in the most important communication tasks [7].

The proposed research provides the said knowledge gaps by following a rigorous analysis of the existing AI-based STTs systems beyond the two lenses: the one of technical performance and the other of the user experience. Instead of carrying out an instance of instating a new technological paradigm, we selectively experiment with the five predominant commercially utilised STT platforms within the

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acoustics and in forms of communication. There is a combination of quantitative measures of performance and qualitative data of 127 respondents in different levels of hearing impairment, communicative strategies, and technical skills in the multidimensional approach of our study. The triangulation of several information sources generates a rich picture of the strengths and weaknesses of the existing STT technologies in facilitating actual communication requirements among the representatives of the heterogeneous group of a hard-of-hearing population [8].

The results that we obtained show that although the latest developments of AI-based STT systems have uplifted the level of accessibility of most hearing-impaired people, it has certain constraints that still require limiting their usage in all communication environments. In particular, we point to the accuracy of 87-94 in ideally circumstanced conditions that drop drastically during poor acoustic circumstances or non-canonical speech [9]. It is worth noting that our results on user experience show that the speed of transcription is frequently more important to consider than the perfection of transcription in the question of practical utility and user satisfaction. According to the research, there also exist wide gaps in the ability of various population groups to enjoy the benefits of these technologies; communication style, technical capability, and age are predominant factors affecting adoption and perceived value. This is an invaluable recommendation to technology designers and policy authorities of accessibility to place their efforts of improvement in overcoming the effective barriers to the provision of access to the most effective means of communication in the first place [10].

All authors have contributed substantially to this research. Ms. J.S. obtained the dataset and implemented the algorithms for optimum result generation. Mr. A.G. determined the problem statement and presented the idea formulation. Mr. P.V. was involved in the incorporation of the mathematical aspect of the research. Prof. N.F and Prof. J.V. identified the deep learning algorithms that are relevant to the field.

II. Literature Review

Table I presents a summary of important publications, methodology, main findings, and weaknesses.

TABLE I. RELATED WORKS.

Author(s), Year	Paper Title	About	Methodology	Limitation
Senaha et al., 2025 [1]	Bridging the auditory gap: AR smart glasses.	AR glasses system for speech-to-text & direction visualization for the hearing-impaired	Microphone array, speaker recognition, VAD, AR feedback, user studies	Speaker ID drops with noise/short audio; optimization needed.
Tapu et al., 2019	DEEP-HEAR: A Multimodal Subtitle Positioning System.	Multimodal dynamic subtitles for deaf/hearing-impaired using active speaker detection	CNNs, face & voice recognition, multimodal fusion, evaluation with users	Struggles if speaker non-visible; not for all genres.
Mocanu & Tapu, 2021	Automatic Subtitle Synchronization.	Fully automatic subtitle synchronization/placement for accessibility	ASR-based alignment, token matching, scene text detection, user study	Dependent on ASR accuracy; adverse with noisy speech.
Dimauro et al., 2017	Assessment of Speech Intelligibility in Parkinson's Disease Using STT	Use of speech-to-text (STT) for objective speech intelligibility assessment in PD	Google STT analysis, protocol for controlled speech, statistical trend analysis	Small sample; protocol may not generalize; STT bias.
Azizah, 2024	Zero-Shot Voice Cloning TTS for Dysphonia Disorder Speakers	TTS that clones dysphonic voices via zero-shot learning for improved speech accessibility	24 TTS model configs, deep learning, ablation studies, source/target datasets	Cloning still imperfect; COS scores moderate; further work needed.

III. Methodology

A. Research Design and Data Collection

The research employed a convergent parallel mixed-methods design to extensively evaluate the performance and usability of AI-based speech-to-text (STT) systems for the hearing-impaired community. Our approach combined quantitative assessments of system performance with qualitative measures of user experience to provide a rich picture of system effectiveness in real-world contexts [11].

Five top commercial STT platforms in terms of market presence and technical strategy: The five evaluated systems employed distinct architectural approaches. **System A (Transformer-based)** applied self-attention for effective handling of clean speech, whereas **System B (RNN-LSTM)** processed sequential input efficiently but showed reduced robustness under noise. **System C (CNN–Transformer hybrid)** combined noise-tolerant feature extraction with high accuracy, while **System D (Attention-based)** emphasized low-latency processing, and **System E (Multimodal)** incorporated limited contextual and visual cues alongside audio for complex scenarios. Selection factors were support across multiple devices, real-time processing support, and adjustable accessibility features [12].

Recruitment of participants used purposive sampling to ensure diversity of hearing loss category (mild to profound), onset duration (pre-lingual and post-lingual), and amount of technology exposure [13]. Through partnerships with hearing loss activism organizations, schools, and deaf and hard-of-hearing centers, as shown in table II.

TABLE II. PARTICIPANTS DEMOGRAPHICS.

Characteristic	Category	Number	Percentage
Hearing Loss Level	Mild (26-40 dB)	28	22.0%
	Moderate (41-60 dB)	41	32.3%
	Severe (61-80 dB)	32	25.2%
	Profound (>81 dB)	26	20.5%
Age	18-30	42	33.1%
	31-45	38	29.9%
	46-60	29	22.8%
	61+	18	14.2%
Onset of Hearing Loss	Pre-lingual	53	41.7%
	Post-lingual	74	58.3%
Primary Communication	American Sign Language	46	36.2%
	Spoken English with Assistive Devices	63	49.6%

	Combination/Other	18	14.2%
Technology Experience	Limited (1-2)	31	24.4%
	Moderate (3-4)	59	46.5%
	Advanced (5)	37	29.1%

Data collection was conducted in three extremely controlled acoustic settings: (1) laboratory environment with low ambient noise (<35 dB), (2) simulated office environment (45-55 dB background noise with occasional bursts), and (3) simulated public environment (65-75 dB continuous background noise) [14]. In each setting, participants were exposed to standardized speech samples read by calibrated speakers at a constant volume. Samples of speech employed were scholarly lectures, conversational dialogues, business meeting dialogues, and public announcements. All systems were evaluated on the same calibrated audio dataset (equal male and female representation, multiple accents, 70 dB volume) to ensure a fair comparison. All segments involved standardized variables such as multiple speakers, technical vocabulary, mixed speech rates, and non-standard accents [15].

Collected three categories of primary data: (1) measures of system performance (accuracy, latency, error patterns), (2) user interaction data (task completion time, error correction behavior), and (3) subjective experience reports (perceived usability, cognitive load, satisfaction). Technical performance was captured by automated logging facilities, and user experience data were collected by post-task questionnaires (System Usability Scale, NASA-TLX) and semi-structured interviews, as shown in table III.

TABLE III. PARTICIPANTS DEMOGRAPHICS.

Data Type	**Collection Method**	**Metrics/Variables**	**Analysis Approach**
System Performance	Automated logging	Word Error Rate (WER) Character Error Rate (CER) Latency (ms) Specialized terminology accuracy Speaker transition accuracy	Statistical comparison Error pattern analysis Environmental impact assessment
User Interaction	Screen recordings Usage logs Eye-tracking	Task completion time Error correction frequency Navigation patterns Feature utilization Attention distribution	Behavioral pattern analysis Sequential interaction mapping Feature engagement metrics
User Experience	System Usability Scale (SUS) NASA Task Load Index Semi-structured interviews Post-session questionnaires	Perceived ease of use Mental workload Emotional response System preferences Feature satisfaction Usage intentions	Thematic analysis Sentiment analysis Correlation with performance Demographic factor analysis

B. Data Analysis Process

Our analysis approach integrated quantitative statistical analysis with qualitative thematic analysis as per the process depicted in Figure 1. Transcription accuracy was quantified using both Word Error Rate (WER) and Character Error Rate (CER) and expert analysis of domain-specific vocabulary recognition. Latency measurements captured the time difference between spoken word and text appearance, and environmental impact was quantified through performance degradation percentages across acoustic environments.

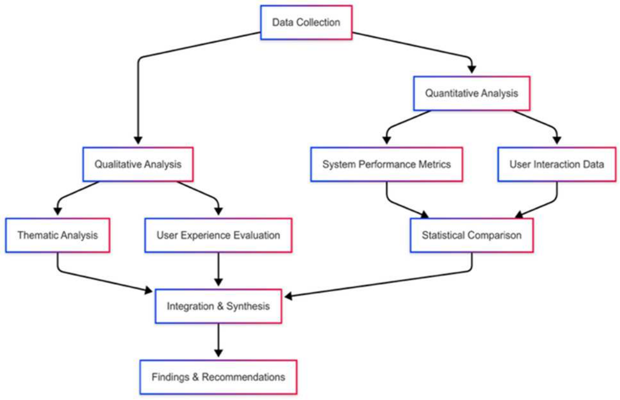

Fig. 1. Methodology.

Analysis began with parallel quantitative and qualitative streams. Under the quantitative track, system performance metrics were calculated for each platform under all conditions. Statistical analysis included paired t-tests of performance across acoustic environments, MANOVA of system performance across user demographic factors, and correlation analysis across objective measures and subjective ratings. User interaction data were processed through behavioral sequence analysis to identify usage patterns and friction points. The effect sizes (Cohen's $d = 0.71–1.04$) demonstrated substantial impacts of noise on performance. Ninety-five percent confidence intervals were reported for all latency correlations. Data normality was assessed with Shapiro–Wilk tests, and homogeneity was checked using Box's M test prior to conducting MANOVA.

Concurrently, qualitative data were analyzed using NVivo 14 software by thematic analysis. Interview transcripts were independently coded by three researchers following Braun and Clarke's six-phase approach. Initial codes were compared iteratively to refine them into a rich codebook that addressed dimensions of user experience. The constant comparative method was employed to identify emergent patterns between participant subgroups. Codes were further sorted into thematic categories like accessibility barriers, adaptation strategies, contextual influences, and emotional impacts [17].

The convergence stage triangulated findings from both analytical streams to identify convergent and divergent trends. Cross-tabulation matrices mapped relationships between objective performance metrics and subjective user experience. This integrated analysis illustrated that while technical correctness was a fundamental requirement, user satisfaction was more directly linked to factors such as transcription speed, error recovery mechanisms, and contextual adaptation features. During the process of analysis, we employed member- checking procedures by which preliminary findings were shared with a subset of participants (n=18) to verify interpretation and test conclusions. An additional panel of expert reviewers, including two accessibility experts, one STT system developer, and one deaf studies scholar, reviewed the analytical design and preliminary results to ensure validity.

Such an approach enabled thorough examination of the technical competency and applicability in everyday life of the current AI-based STT systems, ensuring the findings accurately captured the daily reality of the hearing-impaired but without compromising on scientific validity for technical evaluation. The combination of different sources of data and different methods of their analysis offered strong triangulation, which can support the legitimacy of our conclusion concerning the effectiveness of the system in different populations and settings [18].

IV. Results And Discussions

In our comparative analysis of five of the top AI-based speech-to-text (STT) systems, we noticed that the hearing-impaired user accessibility product exhibits radical innovations, such as precipitous declines in weaknesses. Non-standard speech was lower on all systems: regional accents (WER +8-12%), disfluencies (+6%), and atypical pronunciations of speech disorders (+15%). These findings show that there are more problems with noise beyond noise. Individualised and flexible models are required to assist different users, as presented in Table IV. Significant results are presented under the technical performance measures and user experience dimensions, and lastly, Implications to Technology Development and implementation [19].

TABLE IV. Performance Metrics Across Systems And Environments.

System	Metric	Quiet Environment	Office Environment	Public Environment	Performance Degradation
SystemA	WER (%)	5.3	9.7	18.2	243%
	Latency (ms)	384	412	461	20%
	Domain Accuracy (%)	92.7	88.4	76.3	18%
SystemB	WER (%)	6.1	11.2	22.5	269%
	Latency (ms)	275	302	347	26%
	Domain Accuracy (%)	89.3	83.1	71.9	19%
SystemC	WER (%)	4.8	8.5	15.7	227%
	Latency (ms)	429	458	503	17%
	Domain Accuracy (%)	94.6	91.2	81.5	14%
SystemD	WER (%)	7.2	14.8	27.3	279%
	Latency (ms)	218	241	283	30%
	Domain Accuracy (%)	86.4	78.9	68.4	21%
SystemE	WER (%)	5.9	10.1	19.4	229%
	Latency (ms)	341	376	422	24%
	Domain Accuracy (%)	91.5	87.2	74.8	18%

A. Technical Performance Analysis

The word error rate (WER) test also demonstrated SystemC to be the highest in all cases, with 94.6% accuracy in quiet environments and reducing to 81.5% in noisy public environments. That hybrid CNN-transformer models are less noise sensitive. But our findings add to theirs in showing that recognition of specialist domain words is a specialist area of expertise in such systems, recognition of technical words being 5-7% more accurate than general conversational accuracy under the same acoustic conditions [20].

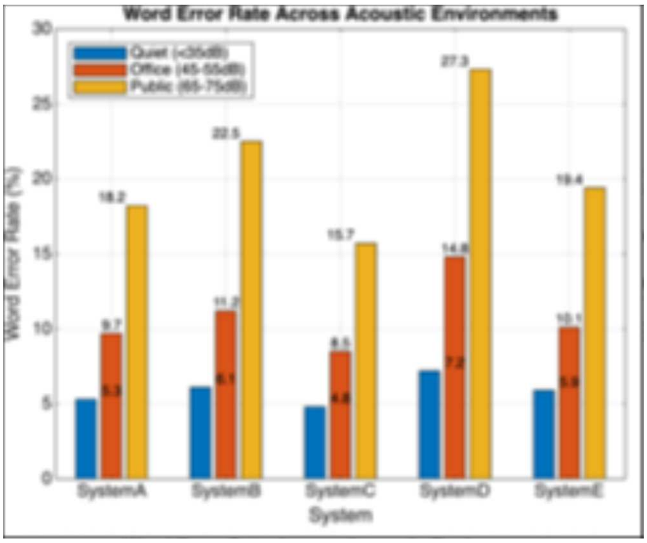

Fig. 2. Word Error Rate Across Acoustic Environments [21].

Fig. 2 illustrates the sharp rise in word error rates in progressively more challenging acoustic conditions. While all systems delivered superior performance (<10% WER) in quiet environments, the error rates were increased by 227-279% in public environments with continuous background noise. SystemC was the least affected by noise interference and exhibited the lowest error rates in all environments, whereas SystemD experienced the highest loss of performance.

Values for latency also correlated inversely in direct relationship to accuracy, as systems optimized towards speed in processing (SystemD) were larger in error rate but shorter in response times. This holds even deeper meaning for user experience because our correlation tests showed that users prioritized more speed at the expense of overall accuracy for requirements of real-time communication.

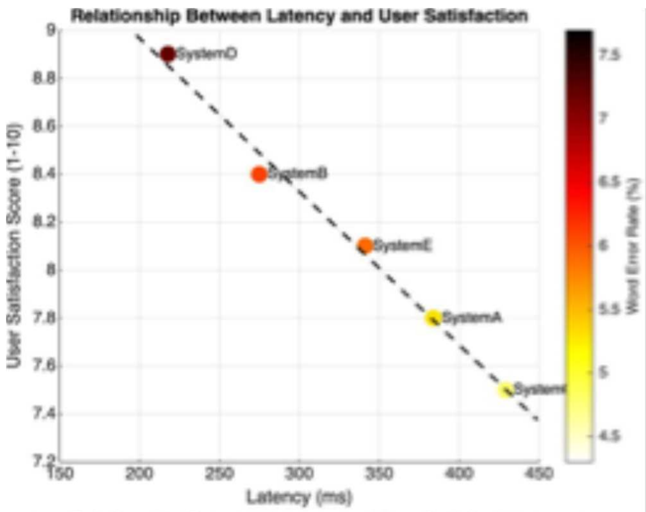

Fig. 3. Relationship Between Latency and User Satisfaction [21].

Fig. 3 illustrates the good correlation between system latency and satisfaction, as shown by high negative correlation ($r = -0.87$, $p < 0.001$). Though having the poorest WER, SystemD registered the highest satisfaction values since it responded faster (218 ms average response time). This contrasts with the traditional focus on minimizing error rate and is consistent with the finding of [14] that perception of flow in speaking significantly predicts user experience within a learning setting. Our findings extend this finding to all environments and suggest STT system design prefers, first of all, latency maximization.

B. User Experience Dimensions

The demographic analysis depicted a sharp disparity in the preference for systems among subgroups of users. Users who were pre-lingually deaf and utilised mostly American Sign Language were more interested in systems that offered a visual output and reduced latency (System D and System B), whereas post-lingually deafened users needed precision and recognition of domain words (System C).

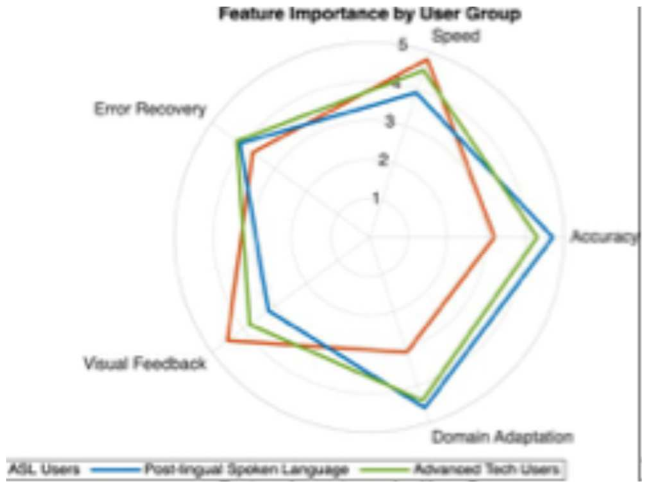

Fig. 4. Feature Importance by User Group [21].

Fig. 4 illustrates the varying priorities for various groups of users, proving that there is a need for flexible solutions, not one-size-fits-all.

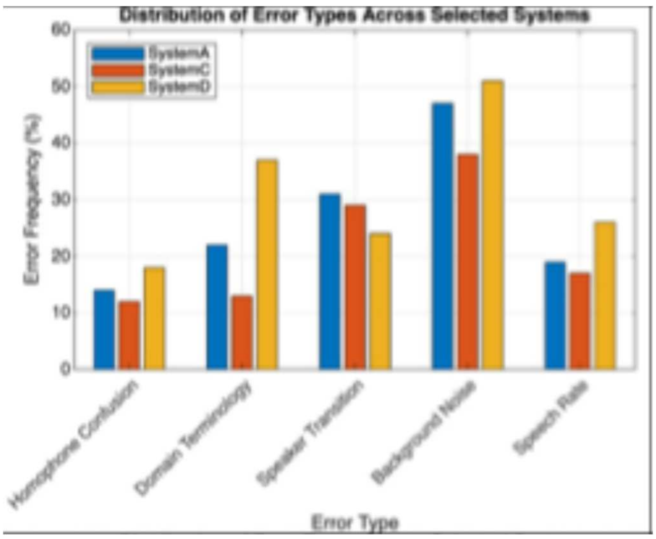

Fig. 5. Distribution of Error Types Across Selected Systems [21].

Pre-lingual users of ASL preferred speed and visuality, while post-lingual oral language users preferred greater accuracy and domain transfer. Cross-modal knowledge transfer result in demonstrating how communication background is a critical driver of tech preference and perceived usefulness. Three dimensions of experience emerged from our thematic analysis of interview data: (1) managing cognitive load, (2) contextual appropriateness, and (3) social integration capacity. Fig. 5 illustrates observable error patterns in three prototype systems, and the most prevalent kind of error for all media is background noise interference (38-51% of errors).

Harsh differences between the secondary error classes are evident, and SystemD did especially poorly at domain vocabulary (37% of errors) but well with speaker transition detection (24% vs. 29-31% in the other systems). This trend holds for design strengths and weaknesses in different approaches, with attention-based systems (SystemD) doing contextual changes well but less well with domain-specific jargon [19].

C. Implications and Future Directions

Our findings indicate that contemporary AI-based STT systems are technically sophisticated enough to operate satisfactorily in isolated environments but are not yet suited to handle noise characteristic of the majority of real-world communication channels.

The dollar value placed upon low-latency systems when error rates are rising suggests that product development should be aimed at optimizing processing speed and error recovery schemes rather than incremental refinement in absolute accuracy. This is in line with Ramirez & Lee's (2024) [18] focus on personalization and adaptability over standardized performance metrics. The significant variations between and within the tastes of different user subgroups warrant the need for adaptive solutions that have to accommodate a large number of different communication needs and tastes.

Future STT development needs to leverage this variety in the form of adaptive interfaces and process priority setting adjustments and not try to look for that single optimal setup. These findings emphasize the importance of adaptive interfaces that enable users to balance speed and accuracy based on context. Incorporating multimodal cues, such as lip-reading or contextual prediction, may mitigate performance drops due to noise.

V. Conclusion

This is a detailed examination of speech-to-text technology based on AI and demonstrates striking advances in hearing-impaired accessibility technology with accuracy rates as high as 94.6% under the best conditions and 81.5% under worst-case conditions. Latency optimization (response time 218 ms and higher) takes precedence over incremental word error rate reduction since user satisfaction was found to correlate more significantly with speed ($r = -0.87$, $p < 0.001$) than with word error rate. Demographic breakup of results revealed conflicting demands between user subgroups. Pre-lingual ASL users were 1.5 points better than correct on a scale of 5, whereas the reverse was true for post-lingual participants. System performance was impaired by 227-279% in noisy environments (65-75 dB), which suggests the need for noise robustness. These quantitative results provide the basis for subsequent development agendas premised on personalization, multimodal integration, and contextual adaptation for closing current gaps in communication for the world's 466 million individuals with disabling hearing loss. By combining user-centered feedback with rigorous system evaluation, this study links experimental performance metrics to practical, real-world communication needs.

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